Paralog classification criteria with gene expression data

Category	Expression	Tissues	Parent comparison	Interpretation
SUPPORTED	Strong (TPM ≥ 0.5)	≥ 2 tissues	Parent expressed, r ≤ 0.8	True paralog; expression independent of parent
ROBUST EXPRESSION	Strong (TPM ≥ 0.5)	≥ 2 tissues	No parent or parent TPM = 0	Good expression; independence cannot be tested
PARENT CORRELATED	Strong (TPM ≥ 0.5)	≥ 2 tissues	Parent expressed, r > 0.8	Co-regulated paralog; mirrors parent expression
LOW EXPRESSION	Weak or narrow	< 2 tissues or TPM < 0.1	Any	Marginal evidence; may be real but lowly expressed
UNSUPPORTED	None	0 tissues	N/A	Zero reads; likely artifact or pseudogene









